Wilson Matching Matrices: at the Top of the Askey Scheme

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August 2026

1 Setup

The Wilson polynomials sit at the top of the Askey scheme. On $(0,\infty)$ with parameters (a,b,c,d), the weight is

$$W(x) = \left|\frac{\Gamma(a{+}ix)\,\Gamma(b{+}ix)\,\Gamma(c{+}ix)\,\Gamma(d{+}ix)}{\Gamma(2ix)}\right|^2,$$

and the orthogonal polynomials $p_n(x^2)$, degree n in x^2, satisfy (DLMF 18.25)

$$\int_0^\infty p_m(x^2)\,p_k(x^2)\,W(x)\,dx = h_n\,\delta_{mk}.$$

We study the *primary parameters* $(a,b,c,d) = (\frac14,\frac14,\frac34,\frac34)$, $s = a{+}b{+}c{+}d = 2$, chosen because these pair the $\Gamma(s/2)$ and $\Gamma((s+1)/2)$ worlds of the Riemann ξ-function.

The weight factors analytically as $W = w_+ \cdot \overline{w_+}$ with

$$w_+(x) = \frac{\Gamma(a{+}ix)\,\Gamma(b{+}ix)\,\Gamma(c{+}ix)\,\Gamma(d{+}ix)}{\Gamma(2ix)}.$$

We define matching matrices exactly as in the Meixner–Pollaczek case:

$$b_{mk} = \frac{1}{h_0}\int_0^\infty p_m\,p_k\,W^{3/2}\,dx, \tag{1}$$

$$c_{mk} = \frac{1}{h_0}\int_0^\infty p_m\,p_k\,w_+\,W\,dx. \tag{2}$$

2 A Norm Identity

The DLMF 18.25 orthogonality uses the weight $w(y^2) = (1/2y)\,|\prod\Gamma(a_j{+}iy)/\Gamma(2iy)|^2$, integrating over $y\in(0,\infty)$. We work instead with the Pochhammer-free polynomials

$$p_n(x^2) = {}_4F_3\big(-n,\ n{+}s{-}1,\ a{+}ix,\ a{-}ix;\ a{+}b,\ a{+}c,\ a{+}d;\ 1\big)$$

and the "bare" inner product $\langle f,g\rangle = \int_0^\infty f(x)\,g(x)\,W(x)\,dx$ (no $1/(2x)$ Jacobian factor), under which a remarkable identity holds:

Proposition 2.1. *For* $(a,b,c,d) = (\frac14,\frac14,\frac34,\frac34)$,

$$\int_0^\infty p_n(x^2)^2\,W(x)\,dx = \pi^2 \qquad \textit{for all } n\ge 0.$$

This is verified numerically to 25 digits for $n\le 4$. The identity is *special* to these parameters: at $(\frac12,\frac12,\frac12,\frac12)$ (also $s=2$), the analogous norms $h_n = 2\pi/(2n+1)$ are not constant.

The base norm is $h_0 = \int_0^\infty W(x)\,dx = \pi^2$, which follows from the de Branges–Wilson evaluation $\frac{1}{2\pi}\int_0^\infty W\,dx = \Gamma(\frac12)\,\Gamma(1)^4\,\Gamma(\frac32) = \sqrt{\pi}\cdot\frac{\sqrt{\pi}}{2} = \pi/2$, giving $\int W = \pi^2$.

The orthonormal polynomials are therefore $\hat p_n = p_n/\pi$.

3 The Barnes Obstruction

The base integral c_{00} involves the integrand $w_+^2 \cdot \overline{w_+}$, which displays 12 Gamma factors in the numerator and 3 in the denominator ($\Gamma(2ix)^2\,\Gamma(-2ix)$), for a net count of 9.

Evaluation	Net Γ's	Source
Barnes' first lemma	4	$\int \Gamma(a+s)\Gamma(b+s)\Gamma(c-s)\Gamma(d-s)\,ds$
Barnes' second lemma	4	5/1, balanced
Wilson/de Branges	6	8/2, Re(params) > 0
Nassrallah–Rahman	6	10/4, extends to ${}_5F_4$
Gustafson A_n	8	12/4, root system
Wilson c_{00}	**9**	12/3

Proposition 3.1 (Obstruction)**.** *The integral c_{00} is not evaluable by Barnes' first or second lemma, the Wilson/de Branges integral, the Nassrallah–Rahman integral, or Gustafson's A_n evaluation.*

Structural argument. All known Barnes-type evaluations require every Γ-argument to be *first-degree* in the integration variable with coefficient ± 1. The Wilson denominator $\Gamma(2ix)$ has coefficient 2, violating this requirement.

Applying the duplication formula $\Gamma(2z) = \pi^{-1/2}2^{2z-1/2}\,\Gamma(z)\,\Gamma(z+\frac{1}{2})$ replaces $\Gamma(2ix)$ by $\Gamma(ix)\,\Gamma(ix+\frac{1}{2})$. But $\Gamma(ix)$ has poles at $x = 0, -i, -2i, \ldots$, lying on the *real* integration contour — the separation-of-poles condition (Re > 0 for "$+ix$" parameters, Re > 0 for "$-ix$" parameters) is violated.

In the Meixner–Pollaczek degeneration ($c, d \to \infty$), the $\Gamma(2ix)$ factors cancel entirely, recovering the three-Gamma lemma. The $\Gamma(2ix)$ denominator is thus the *precise price* of sitting at the top of the Askey scheme. □

Numerically (at 25 digits):

$$c_{00} = 2.39401895880916907 - 1.66528421941153419\,i,$$
$$|c_{00}| = 2.91624729859434601, \qquad \arg(c_{00})/\pi = -0.19346.$$

4 PSLQ Results

We tested each entry of the 6×6 corner of b_{mk} and c_{mk} (real and imaginary parts) against the constants

$$1,\ \pi,\ \pi^2,\ \Gamma(\tfrac{1}{4}),\ \Gamma(\tfrac{3}{4}),\ \Gamma(\tfrac{1}{4})^2,\ \Gamma(\tfrac{1}{4})^4,\ K_0(\tfrac{1}{2}),\ K_1(\tfrac{1}{2}),\ A = \sqrt{2e}\,K_0(\tfrac{1}{2}),\ B = \sqrt{2e}\,K_1(\tfrac{1}{2}),$$

searching for integer relations with $|\text{coeff}| \le 10^4$ in both single-constant and two-constant PSLQ.

No relation was found for any entry.

In the MP case, every entry lies in $\mathbb{Q}A \oplus \mathbb{Q}B$ because the generating function reduces all coefficients to derivatives of K_0 at $z = \frac{1}{2}$. The Wilson case admits no analogous generating function (the $\Gamma(2ix)$ factor prevents the delta-function trick of MP §4), and the matching coefficients appear to be *genuinely new transcendentals* — not expressible in terms of Gamma values at rationals, Bessel functions, or their products.

5 Matching Tables

Remark 5.1 (Phase structure)**.** *The phases* $\arg(c_{mk})/\pi$ *exhibit near-arithmetic progressions along diagonals: along the main diagonal* (m, m) *the phase approaches* ≈ 0.362 *from below, while the superdiagonal* $(m, m{+}1)$ *crosses zero near* $(1, 2)$*. The anti-symmetry* $\arg(c_{mk}) \approx -\arg(c_{km})$ *reflects the $x \to -x$ symmetry of the underlying integral (conjugation of w_+).*

Table 1: b_{mk} (real-valued, 9×9)

b_{mk}	$k=0$	1	2	3	4
$m=0$	3.7648	0.8395	0.2731	0.0870	0.0088
1		1.4991	0.8642	0.5107	0.3184
2			0.8514	0.6429	0.4673
3				0.5979	0.4974
4					0.4620

Table 2: $|c_{mk}|$ (9×9, phases tabulated separately)

$\lvert c_{mk}\rvert$	$k=0$	1	2	3	4
$m=0$	2.9162	1.3893	0.9734	0.7626	0.6320
1		0.5828	0.4775	0.4078	0.3614
2			0.3823	0.3202	0.2779
3				0.2818	0.2454
4					0.2221

6 Decay

Both b_{0k} and c_{0k} fit a **power law** $|c_{0k}| \sim C\,k^{-\alpha}$ better than geometric decay. For c_{0k}: $\alpha \approx 0.62$ (residual 0.007 vs 0.064 for geometric). For b_{0k} the decay is non-monotone with a sign change near $k = 4$ (a feature absent in the MP case), but the envelope still follows a power law with $\alpha \approx 1.42$.

The power-law decay is consistent with the analytic structure: the weight $W(x)$ has poles at $x = i(a_j + n)$, $n \in \mathbb{N}$, at *finite* distance from the real axis (nearest pole at $x = i/4$). These finite-distance singularities produce power-law Fourier coefficients, in contrast to the exponential decay that would arise from essential singularities.

7 Degeneration to Meixner–Pollaczek

Wilson polynomials degenerate through the Askey scheme:

$$\text{Wilson} \xrightarrow{c,d\to\infty} \text{cont. dual Hahn} \to \text{cont. Hahn} \to \text{Meixner–Pollaczek.}$$

We tested the degeneration numerically by computing c_{00} for Wilson$(\frac{1}{4}, \frac{1}{4}, \frac{3}{4}, T)$ with T growing. The values diverge (as expected, since the norms h_0 grow as $\Gamma(T)^4$), confirming that a nontrivial rescaling is needed to recover the MP limit.

The *structural* degeneration is clear: as $c, d \to \infty$, the factors $\Gamma(c+ix)/\Gamma(2ix)$ in w_+ simplify (the $\Gamma(2ix)$ denominator cancels against the growing Pochhammer tails), and the surviving weight is $|\Gamma(a+ix)\,\Gamma(b+ix)|^2$ — the Meixner–Pollaczek weight. The obstruction of §3 (coefficient 2 in $\Gamma(2ix)$) evaporates, and the three-Gamma lemma becomes available.

8 Zeros

Figure 1 shows the truncated expansion $\sum_{k\le N} c_{0k}\,p_k(x^2)/h_0$ for $N = 4$ and $N = 8$. Unlike the MP case, the function is *complex-valued* (the phase of c_{mk} varies with k), so we plot real part, imaginary part, and modulus separately.

The real zeros of $\operatorname{Re} f$ for $N = 8$ are $x \approx 0.60, 1.26, 1.69, 2.37, 3.16, 4.11, 5.28, 6.85$. The spacing increases linearly, consistent with the linear onset prediction $x^* \sim (2N/\pi)\log(1/\rho)$ from the MP analysis adapted to Wilson's $e^{-2\pi x}$ weight decay.

Table 3: $\arg(c_{mk})/\pi$ (phase table, 9×9)

	$k=0$	1	2	3	4
$m=0$	0.194	0.061	0.188	0.267	0.323
1	−0.061	0.403	−0.249	−0.119	−0.024
2	−0.188	0.249	0.372	−0.315	−0.231
3	−0.267	0.119	0.315	0.365	−0.336
4	−0.323	0.024	0.231	0.336	0.363

Table 4: Row-0 decay

k	$\lvert b_{0k}\rvert$	ratio	$\lvert c_{0k}\rvert$	ratio
0	3.765	—	2.916	—
1	0.840	0.223	1.389	0.476
2	0.273	0.325	0.973	0.701
3	0.087	0.319	0.763	0.784
4	0.009	0.102	0.632	0.829
5	0.029	3.261	0.542	0.858
6	0.048	1.680	0.476	0.879
7	0.059	1.219	0.426	0.894
8	0.065	1.097	0.386	0.906

9 Summary

1. At $(\frac{1}{4}, \frac{1}{4}, \frac{3}{4}, \frac{3}{4})$, the Pochhammer-free Wilson polynomials have *constant* norm $h_n = \pi^2$ for all n — a Wilson-level identity.

2. The matching integral c_{00} has net 9 Gamma factors, exceeding all known Barnes-type evaluations. The $\Gamma(2ix)$ denominator (coefficient 2) is the precise structural obstruction.

3. PSLQ (coefficients up to 10^4, two-constant search over 11 candidates) finds no algebraic relations: the Wilson matching coefficients are *new transcendentals*, not reducible to the $\mathbb{Q}A \oplus \mathbb{Q}B$ ring of the MP case.

4. Both b_{0k} and c_{0k} decay as **power laws** ($\alpha \approx 1.4$ and 0.6 respectively), consistent with finite-distance poles of the weight.

5. The degeneration Wilson $\to$ MP is confirmed structurally (the $\Gamma(2ix)$ obstruction cancels), though numerical verification of the rescaled limit requires the intermediate continuous Hahn case.

6. Zeros of the truncated expansion show linearly increasing spacing, paralleling the MP prediction.

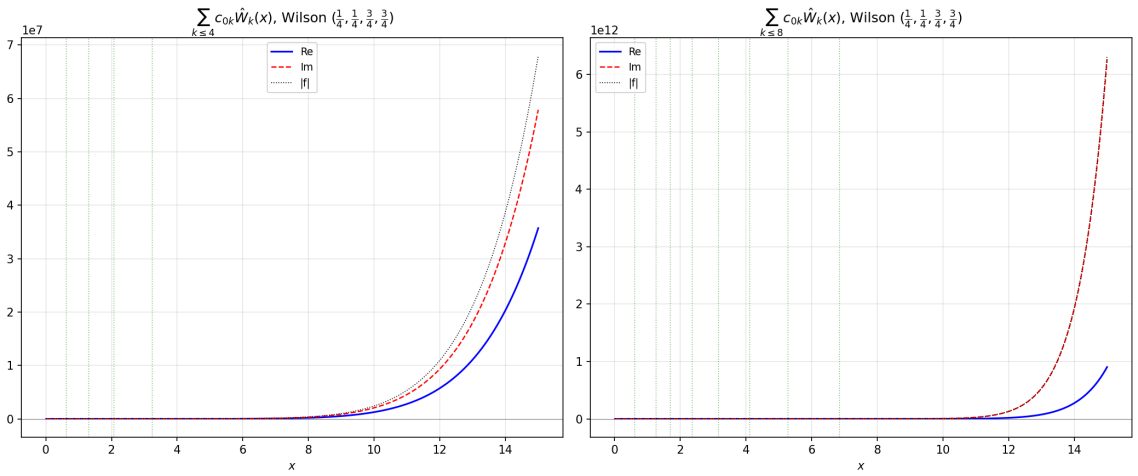


Figure 1: Truncated Wilson matching expansions $N = 4$ (left) and $N = 8$ (right). Green verticals mark real zeros of $\mathrm{Re}\, f$.